

Genetic Analysis Report

Report ID:	SEQ-4D00FF5B0CDC
Generated:	2025-12-04 18:43:19
Patient Name:	Genetic Risk Analysis Report - MODERATE RISK

Analysis Accuracy

95.1%

Patient Information

Analysis performed on file DNA-9EF3B028C777. Processing completed using GENO AI v2.0 Elastic Net Logistic Regression model. Model trained on gene expression data with 94.2% cross-validation accuracy. Risk probability calculated: 40.0%

Variant Information

Gene Expression Analysis Complete • Analyzed Genes: 237 • Mean Expression Level: 45.01 • Expression Variance: 50223.67 • High Expression Markers: 58 • Risk Assessment: MODERATE RISK • Confidence: 40.0%

Analysis Results

{"report": {"sample_id": "SEQ-4D00FF5B0CDC", "laboratory_user_id": "F55055", "generated_at": "2025-12-04T15:43:19.411784Z", "model_name": "geno_enet_pipeline.pkl", "model_version": "2.0", "total_genes_in_model": 237, "genes_analyzed": 237, "risk_level": "LOW", "risk_score_percent": 0.01, "risk_score_raw": 6.139023796168317e-05, "risk_threshold": 75.0, "summary_title": "Genetic Risk Analysis Report - LOW RISK", "summary_text": "The genetic analysis indicates a LOW addiction risk profile based on the evaluated gene-expression markers. The model analyzed 237 genetic markers and calculated a risk probability of 0.01%. This score is below the 75% threshold, indicating lower risk.", "top_genes": [{"gene": "ENSG00000140264.21", "expression": 2721.0}, {"gene": "ENSG00000159788.20", "expression": 1455.0}, {"gene": "ENSG00000125356.7", "expression": 917.0}, {"gene": "ENSG00000110092.4", "expression": 893.0}, {"gene": "ENSG00000099795.7", "expression": 587.0}, {"gene": "ENSG00000100216.6", "expression": 504.0}, {"gene": "ENSG00000149743.14", "expression": 341.0}, {"gene": "ENSG00000150779.12", "expression": 297.0}, {"gene": "ENSG00000232388.5", "expression": 289.0}, {"gene": "ENSG00000229358.3", "expression": 278.0}], "bottom_genes": [{"gene": "ENSG00000180068.10", "expression": 0.0}, {"gene": "ENSG00000231622.1", "expression": 0.0}, {"gene": "ENSG00000220132.1", "expression": 0.0}, {"gene": "ENSG00000269349.1", "expression": 0.0}, {"gene": "ENSG00000284484.1", "expression": 0.0}], "gene_stats": {"mean_expression": 45.0084, "median_expression": 0.0, "std_expression": 224.5807, "min_expression": 0.0, "max_expression": 2721.0, "genes_above_mean": 24,

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This report was generated automatically from genetic data analysis. For questions or concerns, please contact your healthcare provider.